GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

row 22 - INPUT A BYTES PER ELEMENT

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matA})}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matB})}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{D}\downarrow)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matD}\downarrow)}]$$

row 33 - EU COUNT

$$|\mathrm{EU}| = |\mathrm{XeCore}|_{\mathrm{pXeCU}} \times |\mathrm{EU}|_{\mathrm{pXeCore}} \times |\mathrm{XeCU}|$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$|\mathrm{L2}|_{\mathrm{B}} = \mathrm{L2BankSize}_{\mathrm{MB}} \times |\mathrm{L2Banks}|_{\mathrm{pXeCU}} \times 1024 \times 1024$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\mathrm{MemBW}_{\mathrm{BpClk}}^{\mathrm{max}} = \frac{\mathrm{MemBW}_{\mathrm{GBps}}^{\mathrm{max}}}{\mathrm{f}_{\mathrm{GHz}}^{(\mathrm{GT})}}$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{\mathrm{mMACpClk}\cdot\mathrm{XeCore}}^{(\mathrm{peak})} = \min\left(\frac{4}{\frac{\mathrm{FLOOR}(\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} \times 2, 1)}{2}}, \frac{4}{\frac{\mathrm{FLOOR}(\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} \times 2, 1)}{2}}\right) \times \mathrm{D}_{\mathrm{DPAS}} \times 16 \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 42 - CLKS PER DPAS

$$\mathrm{CLKS}_{\mathrm{DPAS}} = \frac{\mathrm{CLKS}_{\mathrm{DPAS}_{\mathrm{num}}}}{\mathrm{CLKS}_{\mathrm{DPAS}_{\mathrm{den}}}}$$

$$\mathrm{CLKS}_{\mathrm{DPAS}_{\mathrm{num}}} = \mathrm{M}_{\mathrm{pThread}} \times \mathrm{K}_{\mathrm{pThread}} \times \mathrm{N}_{\mathrm{pThread}}$$

$$\mathrm{CLKS}_{\mathrm{DPAS}_{\mathrm{den}}} = \frac{\mathrm{CLKS}_{\mathrm{DPAS}_{\mathrm{den}_{\mathrm{num}}}}}{|\mathrm{EU}|_{\mathrm{pXeCore}}}$$

$$\mathrm{CLKS}_{\mathrm{DPAS}_{\mathrm{den}_{\mathrm{num}}}} = \tau_{\mathrm{mMACpClk}\cdot\mathrm{XeCore}}^{(\mathrm{peak})}$$

row 13 - WAVES

$$N_{\text{waves}} = \text{CEILING}(|\text{Tiles}|_{\text{N}}^{(\text{GPU})}, 1) \times \text{CEILING}(|\text{Tiles}|_{\text{M}}^{(\text{GPU})}, 1)$$

row 14 - TG TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{TG})} = \frac{|\text{Tiles}|_{\text{N}}^{(\text{TG})}{}_{\text{num}}}{|\text{Tiles}|_{\text{N}}^{(\text{TG})}{}_{\text{den}}}$$

$$|\text{Tiles}|_{\text{N}}^{(\text{TG})}{}_{\text{num}} = \text{N}_{\text{dim}}$$

$$|\text{Tiles}|_{\text{N}}^{(\text{TG})}{}_{\text{den}} = \text{W}_{\text{element}}^{(\text{ThreadGroup,realized by multiple MMA iterations})}$$

row 15 - TG TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{TG})} = \frac{\text{M}_{\text{dim}}}{\text{H}_{\text{element}}^{(\text{ThreadGroup})}}$$

row 16 - TG CLUSTER TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{TG Cluster})} = \frac{|\text{Tiles}|_{\text{N}}^{(\text{TG Cluster})}{}_{\text{num}}}{\text{W}_{\text{element}}^{(\text{XeCoreCluster})}}$$

$$|\text{Tiles}|_{\text{N}}^{(\text{TG Cluster})}{}_{\text{num}} = \text{N}_{\text{dim}}$$

row 17 - TG CLUSTER TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{TG Cluster})} = \frac{|\text{Tiles}|_{\text{M}}^{(\text{TG Cluster})}{}_{\text{num}}}{\text{H}_{\text{element}}^{(\text{XeCoreCluster})}}$$

$$|\text{Tiles}|_{\text{M}}^{(\text{TG Cluster})}{}_{\text{num}} = \text{M}_{\text{dim}}$$

row 18 - XECU TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{XeCU})} = \frac{\text{N}_{\text{dim}}}{\text{W}_{\text{element}}^{(\text{XeCUTile})}}$$

row 19 - XECU TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{XeCU})} = \frac{\text{M}_{\text{dim}}}{\text{H}_{\text{element}}^{(\text{XeCUTile})}}$$

row 20 - GPU TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{GPU})} = \frac{|\text{Tiles}|_{\text{N}}^{(\text{GPU})}{}_{\text{num}}}{\text{W}_{\text{XeCU}}^{(\text{GPUTile})}}$$

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})}{}_{\mathrm{num}} = \frac{|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})}{}_{\mathrm{num}_{\mathrm{num}}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}$$

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})}{}_{\mathrm{num}_{\mathrm{num}}} = \mathrm{N}_{\mathrm{dim}}$$

row 21 - GPU TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})} = \frac{|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})}{}_{\mathrm{num}}}{\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})}{}_{\mathrm{num}} = \frac{|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})}{}_{\mathrm{num}_{\mathrm{num}}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}$$

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})}{}_{\mathrm{num}_{\mathrm{num}}} = \mathrm{M}_{\mathrm{dim}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{N}_{\mathrm{pThread}}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{M}_{\mathrm{pThread}}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})} = \mathrm{W}_{\mathrm{thread}}^{(\mathrm{ThreadGroup})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})} = \mathrm{H}_{\mathrm{thread}}^{(\mathrm{ThreadGroup})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})} = \mathrm{W}_{\mathrm{TG,keep\ cluster\ sizeas4}}^{(\mathrm{XeCoreCluster})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})} = \mathrm{H}_{\mathrm{TG,keep\ cluster\ sizeas4}}^{(\mathrm{XeCoreCluster})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} = \mathrm{W}_{\mathrm{TG}}^{(\mathrm{XECUTile})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} = \mathrm{H}_{\mathrm{TG}}^{(\mathrm{XECUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} = \frac{|\mathrm{XeCU}|}{\mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{GPUTile})} = \mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{GPUTile})} = \mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}$$

row 63 - MAT A INPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{K}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})}$$

row 64 - MAT B INPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} = \mathrm{K}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matC,matD})} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{D}\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matD}\downarrow)} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Byte}_{\mathrm{pElement}}^{(\mathrm{matD})}$$

row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF THREAD DIVERGENCE

$$|\mathrm{WorkingSet}|_{\mathrm{20Kclksofthreaddivergence}}^{(\mathrm{KinL2})} = \min(20000 \times \frac{\mathrm{K}_{\mathrm{pThread}}}{\mathrm{CLKS}_{\mathrm{DPAS}}}, \mathrm{K}_{\mathrm{dim}})$$

row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE INSTANCE AND SINGLE WAVE

$$\begin{aligned}\mathrm{TotalRequiredL2Size}_{\mathrm{matB\ alwayshit}}^{\mathrm{1\ instance,\ 1\ wave}} &= \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} \times |\mathrm{WorkingSet}|_{\mathrm{20Kclksofthreaddivergence}}^{(\mathrm{KinL2})} \\ &+ \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} \times |\mathrm{WorkingSet}|_{\mathrm{20Kclksofthreaddivergence}}^{(\mathrm{KinL2})} \\ &+ \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{D}\downarrow)}\end{aligned}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$T_{clk}^{(total)} = \frac{{T_{clk}^{(total)}}_{num}}{\eta_{systolic}}$$

$${T_{clk}^{(total)}}_{num} = \frac{{T_{clk}^{(total)}}_{num_{num}}}{\tau^{(peak)}_{mMACpClk \cdot XeCore}}$$

$${T_{clk}^{(total)}}_{num_{num}} = \frac{{T_{clk}^{(total)}}_{num_{num_{num}}}}{|XeCU|}$$

$${T_{clk}^{(total)}}_{num_{num_{num}}} = \frac{{T_{clk}^{(total)}}_{num_{num_{num_{num}}}}}{|XeCore|_{pXeCU}}$$

$${T_{clk}^{(total)}}_{num_{num_{num_{num}}}} = M_{dim} \times K_{dim} \times N_{dim}$$

row 69 - TOTAL L2 READ B

$$L2Rd_{B}^{(total)} = Size_{B}^{(matA)} \times CEILING(\frac{N_{dim}}{W_{element}^{(ThreadGroup, realized\ by\ multiple\ MMA\ iterations)}}, 1)$$
$$+ Size_{B}^{(matB)} \times CEILING(\frac{M_{dim}}{H_{element}^{(ThreadGroup)}}, 1)$$

row 70 - TOTAL L2 WRITE B

$$L2Wr_{B}^{(total)} = Size_{B}^{(matC,matD)}$$

row 71 - TOTAL L1 READ B

$$L1Rd_{B}^{(total)} = Size_{B}^{(matA)} \times CEILING(\frac{N_{dim}}{W_{element}^{(Thread)}}, 1)$$
$$+ Size_{B}^{(matB)} \times CEILING(\frac{M_{dim}}{H_{element}^{(Thread)}}, 1)$$

row 72 - TOTAL L1 WRITE B

$$L1Wr_{B}^{(total)} = Size_{B}^{(matC,matD)}$$

row 74 - L2 READ B XECORE CLK

$$L2RdB_{p(Clk \cdot XeCore)} = \frac{{L2RdB_{p(Clk \cdot XeCore)}}_{num}}{T_{clk}^{(total)}}$$

$$\mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}}} = \frac{\mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}_{\mathrm{num}}}}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{pXeCU}}}$$

$$\mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}_{\mathrm{num}}}} = \mathrm{L2Rd}_{\mathrm{B}}^{(\mathrm{total})}$$

row 75 - L2 WRITE B XECORE CLK

$$\mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}} = \frac{\mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

$$\mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}}} = \frac{\mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}_{\mathrm{num}}}}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{pXeCU}}}$$

$$\mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}_{\mathrm{num}}}} = \mathrm{L2Wr}_{\mathrm{B}}^{(\mathrm{total})}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\begin{aligned}\mathrm{L2RdWrB}_{\mathrm{p(Clk \cdot XeCore)}} &= \mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}} \\ &+ \mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}}\end{aligned}$$

row 77 - L1 READ B XECORE CLK

$$\mathrm{L1RdB}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}} \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 78 - L1 WRITE B XECORECLK

$$\mathrm{L1WrB}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}} \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 79 - L1 READ B EU CLK

$$\mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}} = \frac{\mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

$$\mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}}} = \frac{\mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}_{\mathrm{num}}}}}{|\mathrm{EU}|}$$

$$\mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}_{\mathrm{num}}}} = \mathrm{L1Rd}_{\mathrm{B}}^{(\mathrm{total})}$$

row 80 - L1 WRITE B EU CLK

$$\mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}} = \frac{\mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

$$\mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}}} = \frac{\mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}_{\mathrm{num}}}}}{|\mathrm{EU}|}$$

$$\mathrm{L1WrB}_{\mathrm{p(Clk \cdot EU)}_{\mathrm{num}_{\mathrm{num}}}} = \mathrm{L1Wr}_{\mathrm{B}}^{(\mathrm{total})}$$

row 83 - L2 READ MAX B XECORE CLK

$$\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}} = \frac{\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}}}}{|\mathrm{XeCore}|_{\mathrm{pXeCU}}}$$

$$\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}_{\mathrm{num}}} = |\mathrm{XeCore}|_{\mathrm{pXeCU}} \times 64$$

row 84 - L2 WRITE MAX B XECORE CLK

$$\mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$\mathrm{L2RdWrB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}} = \mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{\mathrm{L2RdB/p(Clk \cdot XeCore)}} = \frac{\eta_{\mathrm{L2RdB/p(Clk \cdot XeCore)}_{\mathrm{num}}}}{\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}}}$$

$$\eta_{\mathrm{L2RdB/p(Clk \cdot XeCore)}_{\mathrm{num}}} = \mathrm{L2RdB}_{\mathrm{p(Clk \cdot XeCore)}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{\mathrm{L2WrB/p(Clk \cdot XeCore)}} = \frac{\eta_{\mathrm{L2WrB/p(Clk \cdot XeCore)}_{\mathrm{num}}}}{\mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}}}$$

$$\eta_{\mathrm{L2WrB/p(Clk \cdot XeCore)}_{\mathrm{num}}} = \mathrm{L2WrB}_{\mathrm{p(Clk \cdot XeCore)}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{\mathrm{L2RdWrB/p(Clk \cdot XeCore)}} = \frac{\eta_{\mathrm{L2RdWrB/p(Clk \cdot XeCore)}_{\mathrm{num}}}}{\mathrm{L2RdWrB}^{(\mathrm{max})}_{\mathrm{p(Clk \cdot XeCore)}}}$$

$$\eta_{\mathrm{L2RdWrB/p(Clk \cdot XeCore)}_{\mathrm{num}}} = \mathrm{L2RdWrB}_{\mathrm{p(Clk \cdot XeCore)}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{\mathrm{L1RdB/p(Clk \cdot EU)}} = \frac{\eta_{\mathrm{L1RdB/p(Clk \cdot EU)}_{\mathrm{num}}}}{\mathrm{L1RdBW}^{\mathrm{max}}_{\mathrm{BpClk \cdot EU}}}$$

$$\eta_{\mathrm{L1RdB/p(Clk \cdot EU)}_{\mathrm{num}}} = \mathrm{L1RdB}_{\mathrm{p(Clk \cdot EU)}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{\mathrm{L1WrB/p(Clk\cdot EU)}} = \frac{\eta_{\mathrm{L1WrB/p(Clk\cdot EU)}_{\mathrm{num}}}}{\mathrm{L1WrBW}^{\mathrm{max}}_{\mathrm{BpClk\cdot EU}}}$$

$$\eta_{\mathrm{L1WrB/p(Clk\cdot EU)}_{\mathrm{num}}} = \mathrm{L1WrB}_{\mathrm{p(Clk\cdot EU)}}$$

row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING DATA SET PCT

$$\mathrm{L2HitRate}_{\mathrm{random\ WS\ access}} = \min\left(\frac{|\mathrm{L2}|_{\mathrm{B}}}{\mathrm{TotalRequiredL2Size}^{\mathrm{1\ instance,\ 1\ wave}}_{\mathrm{matB\ alwayshit}}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$\mathrm{L2MissRate} = 1 - \mathrm{L2HitRate}_{\mathrm{random\ WS\ access}}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$\mathrm{L2RdB}_{\mathrm{total}} = \mathrm{L2Rd}^{\mathrm{(total)}}_{\mathrm{B}}$$

row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE PCT

$$\begin{aligned}\mathrm{P}^{\mathrm{(matA)}}_{\mathrm{L2Hit,after1stwave}} = \mathrm{IF}(&\mathrm{Size}^{\mathrm{(matA)}}_{\mathrm{B}}\\ &+\mathrm{Size}^{\mathrm{(matB)}}_{\mathrm{B}}\\ &+\mathrm{Size}^{\mathrm{(matC,matD)}}_{\mathrm{B}} \leq |\mathrm{L2}|_{\mathrm{B}}, 1.0, \mathrm{IF}(\mathrm{K}_{\mathrm{dim}} > 2 \times |\mathrm{WorkingSet}|^{\mathrm{(KinL2)}}_{\mathrm{20Kclksofthreaddivergence}}, 0.0, 1\\ &-\frac{\mathrm{K}_{\mathrm{dim}} - |\mathrm{WorkingSet}|^{\mathrm{(KinL2)}}_{\mathrm{20Kclksofthreaddivergence}}}{|\mathrm{WorkingSet}|^{\mathrm{(KinL2)}}_{\mathrm{20Kclksofthreaddivergence}}}))\end{aligned}$$

row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE PCT

$$\begin{aligned}\mathrm{P}^{\mathrm{(matB)}}_{\mathrm{L2Hit,after1stwave}} = \mathrm{IF}(&\mathrm{Size}^{\mathrm{(matA)}}_{\mathrm{B}}\\ &+\mathrm{Size}^{\mathrm{(matB)}}_{\mathrm{B}}\\ &+\mathrm{Size}^{\mathrm{(matC,matD)}}_{\mathrm{B}} \leq |\mathrm{L2}|_{\mathrm{B}}, 1.0, 0.0)\end{aligned}$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE PCT

$$\mathrm{P}^{\mathrm{(matA)}}_{\mathrm{L2Miss,after1stwave}} = 1 - \mathrm{P}^{\mathrm{(matA)}}_{\mathrm{L2Hit,after1stwave}}$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE PCT

$$P^{(\mathrm{matB})}_{\mathrm{L2Miss,after1stwave}} = 1 - P^{(\mathrm{matB})}_{\mathrm{L2Hit,after1stwave}}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER COLD CACHE

$$\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}} = \frac{\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}_{\mathrm{num}}}}{2}$$

$$\begin{aligned}\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}_{\mathrm{num}}} = {} & \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \\ & + \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times P^{(\mathrm{matA})}_{\mathrm{L2Miss,after1stwave}} \\ & + \mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \\ & + \mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times P^{(\mathrm{matB})}_{\mathrm{L2Miss,after1stwave}} \\ & + \mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}_{\mathrm{num}_{\mathrm{aux4}}}} \times \mathrm{L2MissRate}\end{aligned}$$

$$\begin{aligned}\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}_{\mathrm{num}_{\mathrm{aux4}}}} = {} & \mathrm{L2RdB}_{\mathrm{total}} \\ & - \mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}_{\mathrm{num}_{\mathrm{aux4}_{\mathrm{aux1}}}}}\end{aligned}$$

$$\begin{aligned}\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}_{\mathrm{num}_{\mathrm{aux4}_{\mathrm{aux1}}}}} = {} & \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \\ & + \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times P^{(\mathrm{matA})}_{\mathrm{L2Miss,after1stwave}} \\ & + \mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \\ & + \mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times P^{(\mathrm{matB})}_{\mathrm{L2Miss,after1stwave}}\end{aligned}$$

row 117 - TOTAL HBM WRITE B

$$\mathrm{MemWrB}^{(\mathrm{total})} = \mathrm{Size}^{(\mathrm{matC,matD})}_{\mathrm{B}}$$

row 118 - TOTAL HBM B

$$\mathrm{MemRdWrB}^{(\mathrm{total})} = \mathrm{MemWrB}^{(\mathrm{total})} + \mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{coldstart}}$$

row 121 - HBM READ B CLK

$$\mathrm{MemRdB_{pClk}} = \frac{\mathrm{MemRdB^{(wave)}_{coldstart}}}{\mathrm{T^{(total)}_{clk}}}$$

row 122 - HBM WRITE B CLK

$$\mathrm{MemWrB_{pClk}} = \frac{\mathrm{MemWrB^{(total)}}}{\mathrm{T^{(total)}_{clk}}}$$

row 123 - HBM TOTAL B CLK

$$\mathrm{MemRdWrB_{pClk}} = \frac{\mathrm{MemRdWrB^{(total)}}}{\mathrm{T^{(total)}_{clk}}}$$

row 126 - HBM BW PCT

$$\eta_{\mathrm{MemBW}} = \frac{\mathrm{MemRdWrB_{pClk}}}{\mathrm{MemBW^{max}_{BpClk}}}$$

row 127 - GTI READ BW PCT

$$\eta_{\mathrm{GTIRdBW}} = \frac{\mathrm{MemRdB_{pClk}}}{\mathrm{GtiRdBW^{max}_{BpClk}}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{\mathrm{GTIWrBW}} = \frac{\mathrm{MemWrB_{pClk}}}{\mathrm{GtiWrBW^{max}_{BpClk}}}$$